

# Introduction to Relational Event Models

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🌐: <https://github.com/martinaboschi/rem-tutorial> • •

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2026 Sunbelt Workshop    —    June 22, 2026



Università  
della  
Svizzera  
italiana

# Agenda



| Section                | Presenter | Time          | Abstract                                                                                                      |
|------------------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------|
| <b>Core REMs</b>       | Melania   | 9:00 - 9:40   | Introduction to REM formulation and inclusion of time-aware – exogenous and endogenous – variables.           |
| <b>REM Inference</b>   | Martina   | 9:45 - 10:20  | Overview of likelihood-based estimation methods.                                                              |
| <b>Practical 1</b>     | Melania   | 10:25 - 10:45 | Simulating REMs and fitting model with linear effects.                                                        |
| <b>Non-Linear REMs</b> | Martina   | 10:50 - 11:30 | Modelling non-linear and time-varying covariate effects while accounting for network and actor heterogeneity. |
| <b>Practical 2</b>     | Melania   | 11:35 - 12:00 | Analysing dynamics of species invasion using REMs beyond linear effect models                                 |

# All our materials at the following link:



<https://martinaboschi.github.io/rem-tutorial/>



**NOTES**

**Formulary, Slides, and Notes**

Supporting materials for the workshops

→



**WORKSHOP 1 - PRACTICAL**

**Introduction to Relational Event Models**

Simulating and fitting REMs

→



**WORKSHOP 2 - PRACTICAL**

**All You Need to Know About Relational Hyper-Event Modeling**

Fitting and interpreting RHEMs

→



**SOFTWARE**

**amore package**

R package

→



**RESOURCE**

**Introduction to Relational Event Modelling**

Tutorial paper

→



# Not sure about the notation or need a quick review?



<https://martinaboschi.github.io/rem-tutorial/>



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→





# During our practicals...



You can either **follow along online** or **download the materials** and run them locally!

<https://martinaboschi.github.io/rem-tutorial/>



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## Practical 1: Simulating Relational Events

This practical session focuses on **simulating relational events** using the `amore` package. If you would like to replicate the code locally, feel free to download the files in the box below:

### Download files

- [Simulating-REM-Data.R](#) — R script
- [Simulating-REM-Data.Rmd](#) — R Markdown source

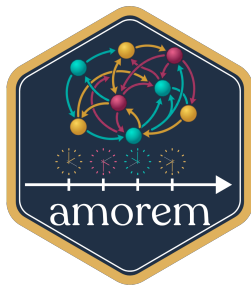
## During our practicals... we will use amore package!



Newly developed R package that implements:

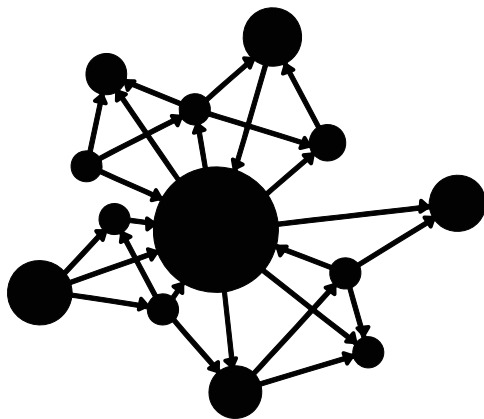
- ▶ simulation of relational event data
- ▶ flexible and computationally efficient inference techniques beyond linear effects

<https://francisorichter.github.io/amorem/>





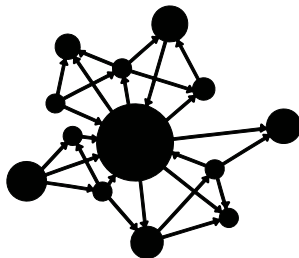
What are some phenomena you would naturally describe as a network?





What are some phenomena you would naturally describe as a network?

- ▶ email communications
- ▶ financial transactions
- ▶ non-native species invasions
- ▶ inter-hospital patient referral



Any of them *static*, for real?



# What is a “longitudinal network”?

**Longitudinal network:** a network where edges have a temporal component:

- ▶ **relational state:** edges appear and persist in time (e.g. friendship)
- ▶ **relational event:** edges are “instantaneous” (e.g. communication)

In this workshop we will focus<sup>1</sup> on

## relational event models

<sup>1</sup>However relational states can be thought of as two relational events:

- ▶ **friendship start:** a relational event marking beginning of state
- ▶ **friendship end:** a relational event marking end of state



# Modeling longitudinal networks

Let  $\{N_{sr}(t)\}_{s \in V^S, r \in V^R, t \geq 0}$  be **longitudinal network**:

- ▶  $N_{sr}(t)$  = number of interactions from  $s$  to  $r$  until time  $t$ .
- ▶  $N_{sr}(0) = 0$

Mathematically, it is always possible to decompose:

$$N_{sr}(t) = \underbrace{\Lambda_{sr}(t)}_{\text{structural part}} + \underbrace{M_{sr}(t)}_{\text{noise part}}$$

If it exists, then the **hazard**

$$\lambda_{sr}(t) = \frac{d\Lambda_{sr}}{dt}(t)$$

is *instantaneous rate* that  $(s, r)$  occurs at time  $t$ .



# What is a hazard?

**Hazard** (also: rate, intensity): “instantaneous probability” that event occurs now.

Mathematically:

$$P((s, r) \text{ occurs } (t, t + \delta t] \mid \text{did not occur until } t) \approx \lambda_{sr}(t)\delta t$$

So, if we know **hazard**:

- ▶ we know **entire probabilistic structure** of event network
- ▶ we can **simulate** relational events:
  - **Gillespie algorithm**: using exponential event times  $\text{Exp}(\lambda_{sr})$ .
  - **tau-leap algorithm**: using small time increments  $\delta t$

# Modeling hazard = modelling network



Let

$$\lambda_{sr}(t) = \underbrace{W_{sr}(t)}_{\text{Is } (s, r) \text{ at risk?}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \underbrace{\exp\{f(x_{sr}(t), t)\}}_{\text{edge-specific risk factors}}$$

where

- ▶  $W_{sr}(t)$ : edge-specific risk indicator  
e.g. a past paper *cannot* cite a future paper.
- ▶  $\lambda_0(t)$ : global risk determinants  
e.g. very few emails sent when it's night at time  $t$ .
- ▶  $f(x_{sr}(t), t)$ : edge-specific risk determinants  
e.g. if  $r$  is a popular person, then  $(s, r)$  more likely.





## How does a relational event network unfold?

Let's consider 5 individuals applying for a job:

- ▶ Each has “work experience”-score  $WE$  given by

0, 3, 5, 9, 11 years

- ▶ They need to get **two references letters** from each other.
- ▶ Person  $s$  contacts person  $r$  for a reference letter with rate,

$$\lambda_{sr}(t) = W_{sr}(t)e^{-1.5|WE_s - WE_r|}.$$

- ▶ Value  $-1.5$  identifies
  - **relevance of WE:** it is not zero
  - **direction of WE:** larger WE difference  $\Rightarrow$  fewer requests
- ▶  $W_{sr}$  has value
  - 1 until person  $s$  has reached out to 2 people,
  - after which it returns to 0.



## Distribution of event times



There are  $5 \times 4 = 20$  requests “waiting” to be asked. Let

$\Delta T_{sr}$  = waiting time until  $s$  asks  $r$  for a letter

The distribution of  $\Delta T_{sr}$  is given as

$$\Delta T_{sr} \sim \text{Exp}(e^{-1.5|WE_s - WE_r|})$$



## When will first request be sent out?

The first request will happen at time  $T_1$ , where

$$T_1 = \min_{sr} \Delta T_{sr}$$

Using property that **minimum of exponentials is exponential**,

$$T_1 \sim \text{Exp} \left( \sum_{s,r} e^{-1.5|WE_s - WE_r|} \approx 0.23 \right)$$





## Which pair of individuals make up first request?

We now know that first event happened at  $t_1$ . But which one?

$E_1$  = event that happened at time  $t_1$ .

Using property that **which exponential is minimum is multinomial**,

$$E_1 \sim \text{Multinomial} \left( \frac{e^{-1.5|WE_1 - WE_2|}}{\sum_{s,r} e^{-1.5|WE_s - WE_r|}}, \dots, \frac{e^{-1.5|WE_5 - WE_4|}}{\sum_{s,r} e^{-1.5|WE_s - WE_r|}} \right)$$





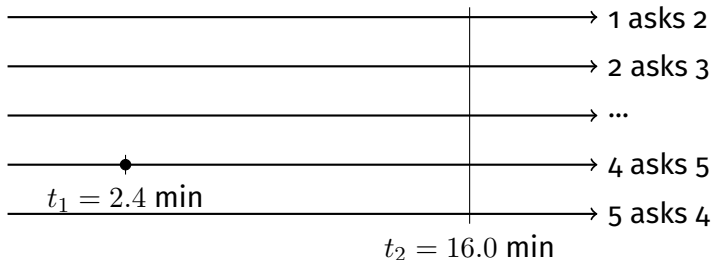
## When will second request be sent out?

The second request will happen at time  $T_2$ , where

$$T_2 = \min_{sr \neq s_1 r_1} \Delta T_{sr}$$

Using [memoryless property of an exponential](#),

$$T_2 \sim T_1 + \text{Exp} \left( \sum_{(s,r) \neq (s_1, r_1)} e^{-1.5|WE_s - WE_r|} \approx 0.18 \right)$$





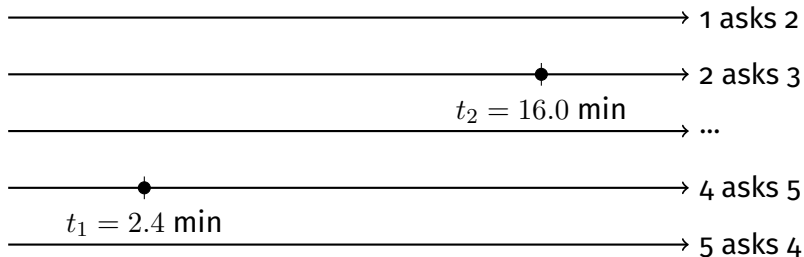
## Which pair of individuals make up second request?

We know second event happened at  $t_2 = 16.0$  min. Which one?

$E_2$  = event that happened at time  $t_2$ .

Using property that **which exponential is minimum is multinomial**,

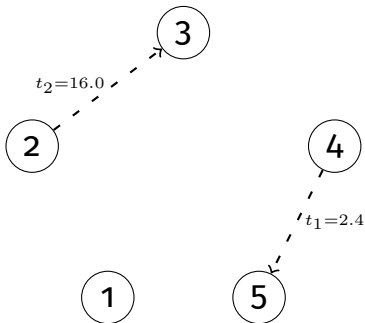
$$E_2 \sim \text{Multinomial} \left( \frac{e^{-1.5|WE_1 - WE_2|}}{0.18}, \dots, \frac{e^{-1.5|WE_5 - WE_4|}}{0.18} \right)$$



## Repeat... to obtain a dynamic network



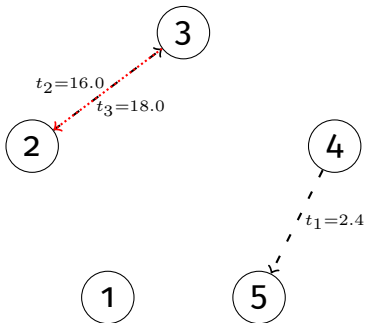
Repeating above procedure another 8 times, results in:



## Repeat... to obtain a dynamic network



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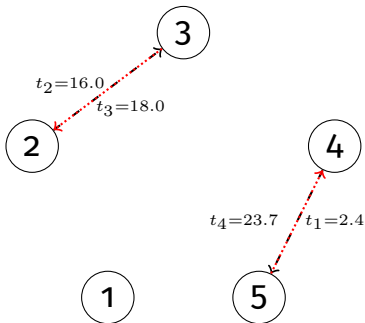




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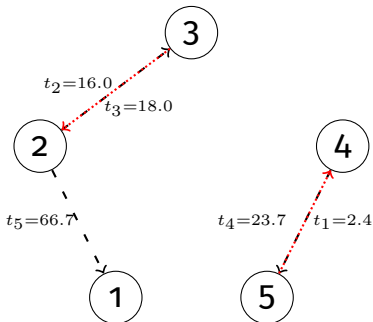
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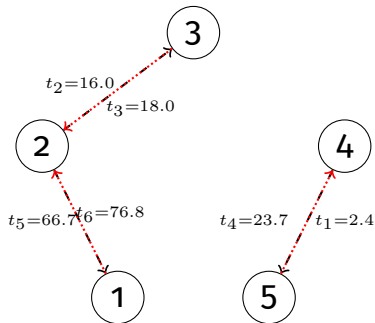
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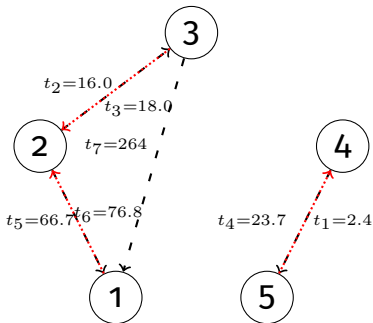
Repeating above procedure another 8 times, results in:





## Repeat... to obtain a dynamic network

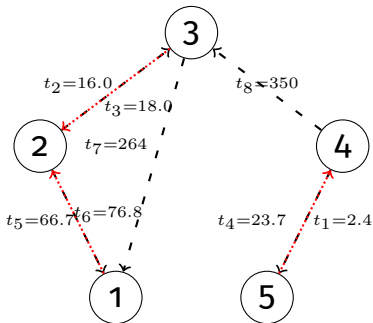
Repeating above procedure another 8 times, results in:



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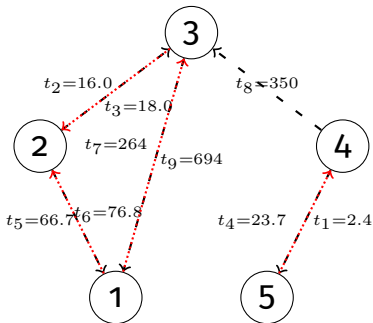
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## Repeat... to obtain a dynamic network



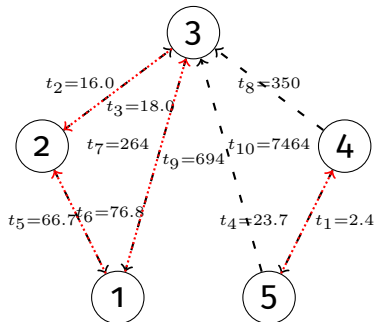
Repeating above procedure another 8 times, results in:





## Repeat... to obtain a dynamic network

Repeating above procedure another 8 times, results in:





## From Data to Answers

Let's assume that someone gave us this dynamic network:

|    | time   | solicitor | receiver | $WE_s$ | $WE_r$ | $WE_{abs-diff}$ |
|----|--------|-----------|----------|--------|--------|-----------------|
| 1  | 2.4    | 4         | 5        | 9      | 11     | 2               |
| 2  | 16.0   | 2         | 3        | 3      | 5      | 2               |
| 3  | 18.0   | 3         | 2        | 5      | 3      | 2               |
| 4  | 23.7   | 5         | 4        | 11     | 9      | 2               |
| 5  | 66.7   | 2         | 1        | 3      | 0      | 3               |
| 6  | 76.8   | 1         | 2        | 0      | 3      | 3               |
| 7  | 264.4  | 3         | 1        | 5      | 0      | 5               |
| 8  | 349.8  | 4         | 3        | 9      | 5      | 4               |
| 9  | 694.1  | 1         | 3        | 0      | 5      | 5               |
| 10 | 7463.6 | 5         | 3        | 11     | 5      | 6               |

Could we determine influence of WE on this dynamic? Yes, by doing **inference**!





# Inference on what?

$$\lambda_{sr}(t) = \underbrace{W_{sr}(t)}_{\text{Is } (s, r) \text{ at risk?}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \underbrace{\exp\{f(x_{sr}(t), t)\}}_{\text{edge-specific risk factors}}$$

We now focus on what the  $x_{sr}(t)$  can be...

Also called:

- ▶ event statistics
- ▶ explanatory variables
- ▶ covariates
- ▶ risk factors
- ▶ drivers



## Exogenous VS Endogenous drivers

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Back to the example...

Work experience similarity  $x_{sr}(t) = |WE_s - WE_r|$  depends on:

- ▶ Work experience score  $WE_s$  which is a node-level attribute, that does not depend on prior interactions → **exogenous** covariate
- ▶ Work experience similarity between  $s$  and  $r$  is dyadic, still exogenous.



## Exogenous VS Endogenous drivers

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- ▶ Work experience similarity between  $s$  and  $r$  is dyadic, still exogenous.
- ▶ Age, gender, occupation, time-zone, distance and any function of them...

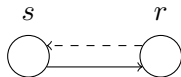
# Exogenous VS Endogenous drivers



But prior events influence future events...

In the reference letters example: *I ask you, you ask me.*

$\leadsto$  Reciprocity (*binary*)  $x_{sr}(t) = \mathbf{1}(\{(r, s, t^*) \mid t^* < t\} \neq \emptyset)$



|    | time   | solicitor | receiver | WE <sub>abs-diff</sub> | REC <sub>binary</sub> |
|----|--------|-----------|----------|------------------------|-----------------------|
| 1  | 2.4    | 4         | 5        | 2                      | 0                     |
| 2  | 16.0   | 2         | 3        | 2                      | 0                     |
| 3  | 18.0   | 3         | 2        | 2                      | 1                     |
| 4  | 23.7   | 5         | 4        | 2                      | 1                     |
| 5  | 66.7   | 2         | 1        | 3                      | 0                     |
| 6  | 76.8   | 1         | 2        | 3                      | 1                     |
| 7  | 264.4  | 3         | 1        | 5                      | 0                     |
| 8  | 349.8  | 4         | 3        | 4                      | 0                     |
| 9  | 694.1  | 1         | 3        | 5                      | 1                     |
| 10 | 7463.6 | 5         | 3        | 6                      | 0                     |

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Depends on prior interactions  $\rightarrow$  **endogenous** covariate

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Depends on prior interactions  $\rightarrow$  **endogenous** covariate

Alternative choice for reciprocity definition:

- ▶  $x_{sr}(t) = \text{number of occurrences of } (r, s) \text{ before } t$
- ▶  $z_{sr}(t) = \begin{cases} t - \tau_{sr}, & \tau_{sr} = \text{time of last event } (r, s) \\ \infty, & \text{otherwise} \end{cases} \rightarrow \text{defined as **internal time**}$
- we transform to  $x_{sr}(t) = \exp(-z_{sr}(t)) \rightsquigarrow$  the closer to zero the further in time, closer to one the opposite.

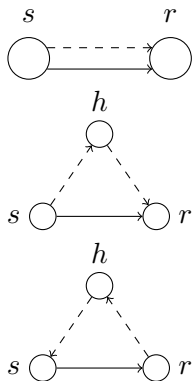
# Exogenous VS Endogenous drivers



But prior events influence future events...

Similar kind of definitions can be used for various endogenous mechanisms:

- ▶ Repetition (*I ask you, I ask you again*)
- ▶ Triadic closure (*I ask you, you ask them, so I ask them*)
- ▶ Cyclic closure (*They ask you, you ask me, so I ask them*)
- ▶ ...

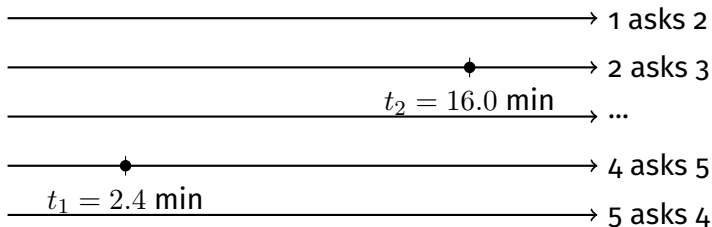




## Versatility isn't cheap...

Endogenous variables powerful for dependence between edges but computationally intensive to compute:

- ▶  $n$  relational events and  $|V^S| = |V^R| = p \rightsquigarrow p^2$  possible interactions
- ▶ for each edge, look at all previous events and check which satisfy the condition of interest... on average  $\frac{n}{2}$
- ▶  $\implies$  Complexity is  $O(n^2 p^2)$  in the worst case, just for dyadic...







## Inference of what?

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{f(x_{sr}(t), t)\}$$

Consider  $f(x_{sr}(t), t) = \beta \cdot x_{sr}(t)$ :

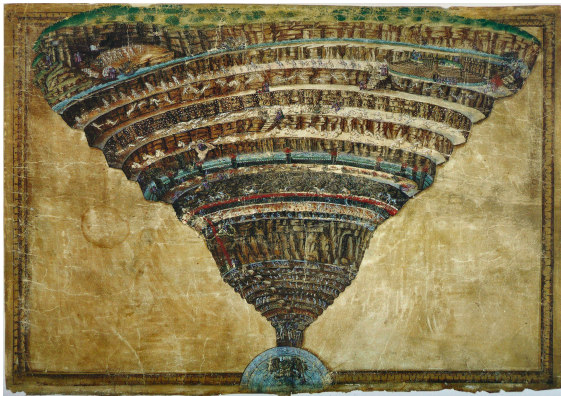
- ▶ Work experience similarity  $x_{sr}(t) = |WE_s - WE_r| \rightsquigarrow \beta < 0$  the greater the difference in work experience, the less likely it is to ask for reference letter
- ▶ Reciprocity  $x_{sr}(t) = \exp(-(t - \tau_{sr})) \rightsquigarrow \beta > 0$  the more recent the email, the more likely it's replied to soon (in principle...)

Estimating effect  $\beta$  given data = inference

# Descending into Dante's Inferno...



*Abandon hope all ye who enter here...*

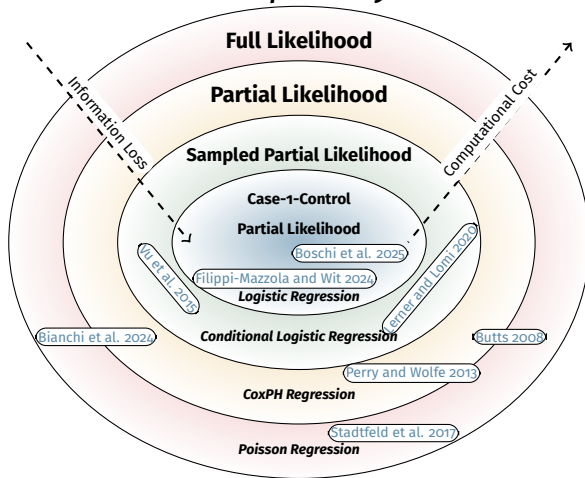


**Fig. 1:** Watts, Barbara J.: “Sandro Botticelli’s Drawings for Dante’s Inferno”



## ...or through rings of Likelihood?

*Maintain* Abandon hope all ye who enter here...



# A quick recap!



**Relational Data:** stream of observed events.

$$E = \{(s_i, r_i, t_i), i = 1, \dots, n\}$$

**Parametric Model:** linear (in log-hazard)... just for now!

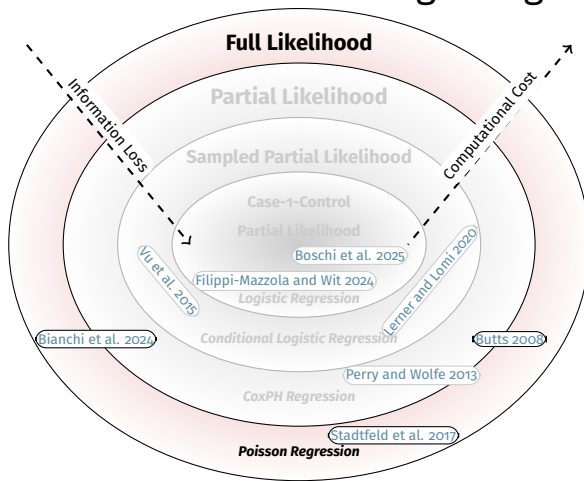
$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{\beta \cdot \mathbf{x}_{sr}(t)\}$$

**Likelihood:** constructed as joint probability of observing **data** under **model**.

**AIM:** estimate  $\beta$  that maximizes **likelihood**.



Let's start from the beginning...





## Full Likelihood: Let's follow the definition...

**Likelihood:** constructed as joint probability of observing **data** under **model**.

Perry and Wolfe [2013]

$$\mathcal{L}(\beta) = p(E; \beta) \quad \underbrace{\quad}_{\text{Multiplication Theorem}} \quad \prod_{i=1}^n \left[ p\left(t_i \mid \underbrace{t_1, (s_1, r_1), \dots, t_{i-1}, (s_{i-1}, r_{i-1})}_{\text{previous events}}; \beta\right) \times \right. \\ \left. p\left((s_i, r_i) \mid t_i, \underbrace{t_1, (s_1, r_1), \dots, t_{i-1}, (s_{i-1}, r_{i-1})}_{\text{previous events}}; \beta\right) \right]$$

events are independent **conditional on history**  
along with covariates, incorporated by  $\mathbb{W} = \{\mathcal{W}_t\}_{t \in \mathbb{R}^+}$

$$= \prod_{i=1}^n p(t_i | \mathcal{W}_{t_i^-}; \beta) p((s_i, r_i) | t_i, \mathcal{W}_{t_i^-}; \beta)$$



$$\mathcal{L}(\beta) = \prod_{i=1}^n p(t_i | \mathcal{W}_{t_i^-}) \times p((s_i, r_i) | t_i, \mathcal{W}_{t_i^-}; \beta)$$

How to account for fact that there are not other dyads observed in  $(t_{i-1}, t_i)$ ?

$$\underbrace{\prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \exp \left[ - \sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u; \beta) du \right]}_{\text{Minimum Generalized Exponential Inter-arrival Time}} \times \underbrace{p((s_i, r_i) | t_i, \mathcal{W}_{t_i^-}; \beta)}_{\text{Multinomial Probability}} \\ \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}$$

## Full Likelihood for... Piecewise Constant Rates (PCR) I



$$\mathcal{L}(\beta) = \underbrace{\prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \exp \left[ - \sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u; \beta) du \right]}_{\text{Minimum Generalized Exponential Inter-arrival Time}} \times \underbrace{\frac{p((s_i, r_i) | t_i, \mathcal{W}_{t_i}^-; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}}_{\text{Multinomial Probability}}$$

What if hazard is assumed to be constant within inter-arrival times?

$$\prod_{i=1}^n \underbrace{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \times \exp \left[ - \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) (t_i - t_{i-1}) \right]}_{\text{Exponential waiting time } \Delta T_i = T_i - T_{i-1} \text{ - Poisson N. of Events in } \Delta T_i} \times \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}$$



## Full Likelihood for... Piecewise Constant Rates (PCR) II



$$\begin{aligned}
 \mathcal{L}(\beta) &\stackrel{\text{PCR}}{=} \prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \times \exp \left[ - \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) (t_i - t_{i-1}) \right] \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)} \\
 &\quad \prod_{i=1}^n \lambda_{s_i r_i}(t_i; \beta) \prod_{(s,r) \in \mathcal{R}_{t_i}} \exp [-\lambda_{sr}(t_i; \beta) (t_i - t_{i-1})]
 \end{aligned}$$

## Full Likelihood for... Piecewise Constant Rates (PCR) III



$$\begin{aligned}
 \mathcal{L}(\beta) &\stackrel{\text{PCR}}{=} \prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \times \exp \left[ - \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) (t_i - t_{i-1}) \right] \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)} \\
 &\quad \prod_{i=1}^n \lambda_{s_i r_i}(t_i; \beta) \prod_{(s,r) \in \mathcal{R}_{t_i}} \exp [-\lambda_{sr}(t_i; \beta) (t_i - t_{i-1})] \\
 &\quad \text{Call } \Delta N_{sr}(t_i) = N_{sr}(t_i) - N_{sr}(t_{i-1}) \\
 &\quad \underbrace{\prod_{i=1}^n \prod_{(s,r) \in \mathcal{R}_{t_i}} (\lambda_{sr}(t_i; \beta))^{\Delta N_{sr}(t_i)} \exp [-\lambda_{sr}(t_i; \beta) (t_i - t_{i-1})]}_{\text{Proportional to likelihood of a Poisson Regression}}
 \end{aligned}$$

# So, which statistical model should we focus on? (Pt. 1)



**Full Log-Likelihood** (Piecewise Constant Rates):

$$\ell(\beta) = \sum_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log(\lambda_{sr}(t_i; \beta)) - \lambda_{sr}(t_i; \beta)(t_i - t_{i-1})$$

**Modeling:**

**Relational Event Model**

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{\beta \cdot \mathbf{x}_{sr}(t)\}$$

**Inference:**

**Poisson Regression**

$$\Delta N_{sr}(t_i) | \mathbf{x}_{sr}(t_i) \stackrel{\text{iid}}{\sim} \text{Poisson}(\mu_{sr}(t_i))$$

$$\log(\mu_{sr}(t_i)) = \beta \cdot \mathbf{x}_{sr}(t_i) + \log(t_i - t_{i-1})$$

# So, which statistical model should we focus on? (Pt. 1)



**Full Log-Likelihood** (Piecewise Constant Rates):

$$\ell(\beta) = \sum_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log(\lambda_{sr}(t_i; \beta)) - \lambda_{sr}(t_i; \beta)(t_i - t_{i-1})$$

**Modeling:**

**Relational Event Model**

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{\beta \cdot \mathbf{x}_{sr}(t)\}$$

**Inference:**

**Poisson Regression**

$$\Delta N_{sr}(t_i) | \mathbf{x}_{sr}(t_i) \stackrel{\text{iid}}{\sim} \text{Poisson}(\mu_{sr}(t_i))$$

$$\log(\mu_{sr}(t_i)) = \beta \cdot \mathbf{x}_{sr}(t_i) + \log(t_i - t_{i-1})$$

Nice... But at what **price**?

1. Strong Assumption;
2. High Computational Cost;



# Full Likelihood in Practice I

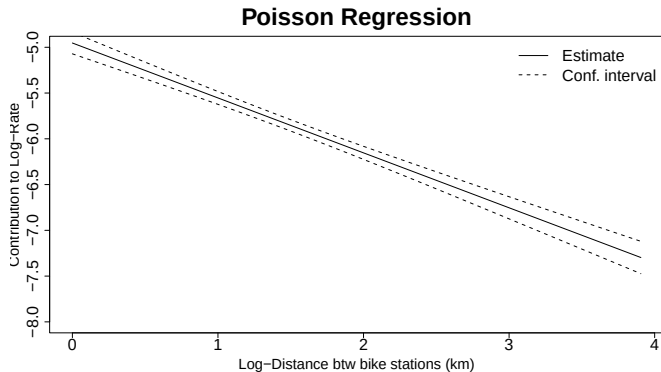
## Full Log-Likelihood (Piecewise Constant Rates):

$$\ell(\beta) = \sum_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log(\lambda_{sr}(t_i; \beta)) - \lambda_{sr}(t_i; \beta)(t_i - t_{i-1})$$

| stratum | event | sender | receiver | time      | dist      | logtm     |
|---------|-------|--------|----------|-----------|-----------|-----------|
| 1       | 1     | 31419  | 31255    | 0.0218023 | 1.6832458 | 0.0218023 |
| 1       | 0     | 31261  | 31071    | 0.0218023 | 1.8376884 | 0.0218023 |
| 1       | 0     | 31937  | 32050    | 0.0218023 | 2.7475273 | 0.0218023 |
| 1       | 0     | 31937  | 31937    | 0.0218023 | 0.0000000 | 0.0218023 |
| 1       | 0     | 31071  | 31023    | 0.0218023 | 1.9742199 | 0.0218023 |
| 1       | 0     | 31937  | 31272    | 0.0218023 | 2.0297259 | 0.0218023 |
| 1       | 0     | 32217  | 31255    | 0.0218023 | 3.8173386 | 0.0218023 |
| 1       | 0     | 32604  | 31966    | 0.0218023 | 2.4291296 | 0.0218023 |
| 1       | 0     | 32050  | 31937    | 0.0218023 | 2.7357951 | 0.0218023 |
| 1       | 0     | 31285  | 31324    | 0.0218023 | 0.5883421 | 0.0218023 |



```
glm(event ~ dist + offset(logtm),
family = poisson(link = "log"), data = poisson_dat)
```





Full Likelihood via Poisson Regression... At what price?

1. Strong Assumption;
2. High Computational Cost;

Can we do better?



Full Likelihood via Poisson Regression... At what price?

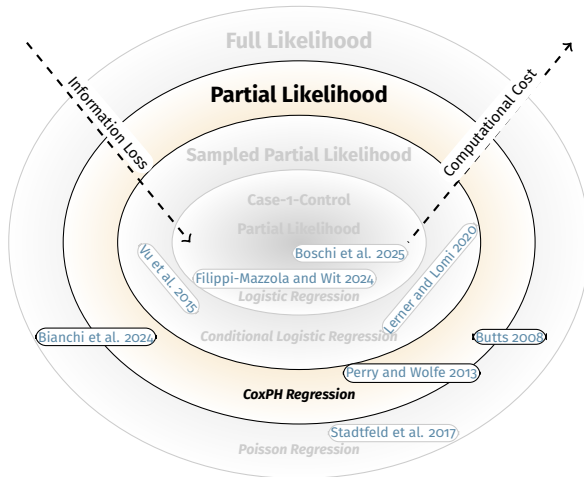
1. Strong Assumption;
2. High Computational Cost;

Can we do better? Let's do one step back:

$$\mathcal{L}(\beta) = \prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \exp \left[ - \sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u; \beta) \mathrm{d}u \right] \times \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}$$



# Partial Likelihood Inference





## Inspired by Cox [1975]: REM's Partial Likelihood I

**Partial likelihood** focuses on **multinomial probability of observing a specific dyad among those at risk, given that an event has occurred**.

$$\mathcal{L}^P(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i | \mathcal{W}_{t_i}^-; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta)} = \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \}}$$



## Inspired by Cox [1975]: REM's Partial Likelihood II

**Partial likelihood** focuses on **multinomial probability of observing a specific dyad among those at risk, given that an event has occurred**.

$$\mathcal{L}^P(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i | \mathcal{W}_{t_i}^-; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta)} = \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \}}$$

### Information Loss:

Leaving  $\lambda_0(t)$  arbitrary  
results in  
**slight loss of information**  
about  $\beta$ .

### Dimensionality reduction:

effective when providing a  
**simplification**  
of full likelihood by  
**eliminating**  
**nuisance parameters.**

### Event Ordering:

Applicable when  
**only order of**  
**events is known,**  
not their exact timing.

## So, which statistical model should we focus on? (Pt. 2)



### Partial Log-likelihood:

$$\ell^P(\beta) = \log \mathcal{L}^P(\beta) = \sum_{i=1}^n \left[ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left( \sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \} \right) \right],$$

**Modeling:**

**Relational Event Model**

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{ \beta \cdot \mathbf{x}_{sr}(t) \}$$

**Inference:**

**CoxPH Regression**

$$\hat{\beta} = \arg \max_{\beta} \ell^P(\beta)$$



# Partial Likelihood in Practice I

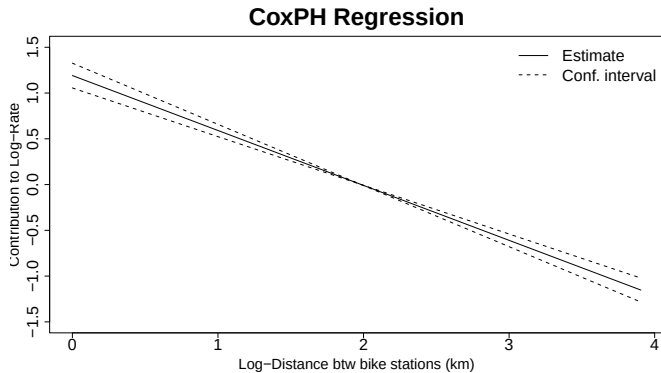
## Partial Log-Likelihood:

$$\ell^P(\beta) = \log \mathcal{L}^P(\beta) = \sum_{i=1}^n \left[ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left( \sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \} \right) \right],$$

| stratum | event | sender | receiver | time      | dist      |
|---------|-------|--------|----------|-----------|-----------|
| 1       | 1     | 31419  | 31255    | 0.0218023 | 1.6832458 |
| 1       | 0     | 31261  | 31071    | 0.0218023 | 1.8376884 |
| 1       | 0     | 31937  | 32050    | 0.0218023 | 2.7475273 |
| 1       | 0     | 31937  | 31937    | 0.0218023 | 0.0000000 |
| 1       | 0     | 31071  | 31023    | 0.0218023 | 1.9742199 |
| 1       | 0     | 31937  | 31272    | 0.0218023 | 2.0297259 |
| 1       | 0     | 32217  | 31255    | 0.0218023 | 3.8173386 |
| 1       | 0     | 32604  | 31966    | 0.0218023 | 2.4291296 |
| 1       | 0     | 32050  | 31937    | 0.0218023 | 2.7357951 |
| 1       | 0     | 31285  | 31324    | 0.0218023 | 0.5883421 |



```
rem(event ~ dist, data = cox_dat, method = "clogit", stratum = "stratum")
```





Full Likelihood via Poisson Regression... At what price?

1. Strong Assumption;
2. High Computational Cost;

Can we do better?



Full Likelihood via Poisson Regression... At what price?

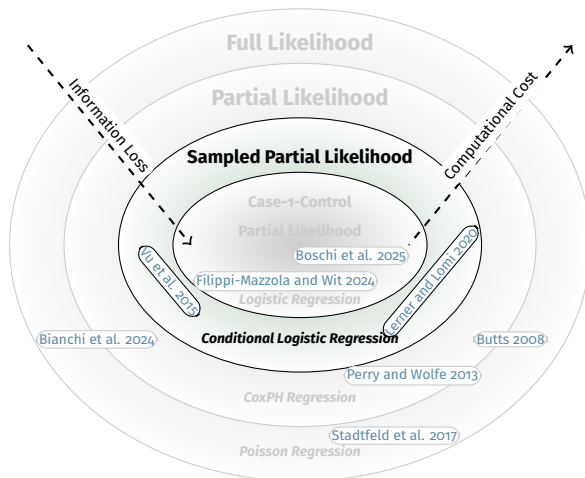
1. Strong Assumption;
2. High Computational Cost;

Can we do better? Let's do one step back:

$$\mathcal{L}^P(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i; \mathcal{W}_{t_i^-}; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \mathcal{W}_{t_i^-}; \beta)} = \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \}}$$



# Sampled Partial Likelihood Inference





# What changes when we sample? I

**Relational Data:** stream of observed events and sampled risk sets.

$$D_m = \{(s_i, r_i, t_i, \mathcal{SR}_{t_i}), |\mathcal{SR}_{t_i}| = m + 1, i = 1, \dots, n\},$$

where  $\mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i}$  is **sampled risk set** at time  $t_i$ .



## What changes when we sample? I

**Relational Data:** stream of observed events and sampled risk sets.

$$D_m = \{(s_i, r_i, t_i, \mathcal{SR}_{t_i}), |\mathcal{SR}_{t_i}| = m + 1, i = 1, \dots, n\},$$

where  $\mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i}$  is **sampled risk set** at time  $t_i$ .

**Updated Filtration:** incorporating history of events and sampling information.

$$\{\mathcal{F}_t\}_{t \geq 0}, \quad \mathcal{F}_t = \mathcal{W}_t \cup \sigma(\mathcal{SR}_{t_i}; t_i \leq t)$$



## What changes when we sample? I

**Relational Data:** stream of observed events and sampled risk sets.

$$D_m = \{(s_i, r_i, t_i, \mathcal{SR}_{t_i}), |\mathcal{SR}_{t_i}| = m + 1, i = 1, \dots, n\},$$

where  $\mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i}$  is **sampled risk set** at time  $t_i$ .

**Updated Filtration:** incorporating history of events and sampling information.

$$\{\mathcal{F}_t\}_{t \geq 0}, \quad \mathcal{F}_t = \mathcal{W}_t \cup \sigma(\mathcal{SR}_{t_i}; t_i \leq t)$$

**Parametric Model:** let's stay linear for the moment...

$$\lambda_{sr, \mathcal{SR}}(t) = \lambda_{sr}(t) \cdot \pi_t(\mathcal{SR} | (s, r)), \quad \pi_t(\mathcal{SR} | (s, r)) = \frac{m}{|\mathcal{R}_t| - 1} \cdot 1_{\{(s, r) \in \mathcal{SR}, \mathcal{SR} \subset \mathcal{R}_t\}}$$

$m$  is the **number of sampled non-events** according to **nested case-control sampling**.



# What changes when we sample? II

## Sampled Partial likelihood:

$$\mathcal{L}^S(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i; \mathcal{F}_{t_i}^-; \beta) \cdot \pi_{t_i}(\mathcal{SR}_{t_i} | (s_i, r_i))}{\sum_{(s,r) \in \mathcal{SR}_{t_i}} \lambda_{sr}(t_i; \mathcal{F}_{t_i}^-; \beta) \cdot \pi_{t_i}(\mathcal{SR}_{t_i} | (s, r))} = \prod_{i=1}^n \frac{\exp\{\beta \cdot x_{s_i r_i}(t_i)\}}{\sum_{(s,r) \in \mathcal{SR}_{t_i}} \exp\{\beta \cdot x_{sr}(t_i)\}}$$

## Look Familiar?

### Conditional logistic regression

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From Wikipedia, the free encyclopedia

**Conditional logistic regression** is an extension of [logistic regression](#) that allows one to account for [stratification](#) and [matching](#). Its main field of application is [observational studies](#) and in particular [epidemiology](#). It was devised in 1978 by [Norman Breslow](#), [Nicholas Day](#), [Katherine Halvorsen](#), [Ross L. Prentice](#) and [C. Sabat](#).<sup>[1]</sup> It is the most flexible and general procedure for matched data.

#### Background [\[ edit \]](#)

Observational studies use [stratification](#) or [matching](#) as a way to control for [confounding](#).

[Logistic regression](#) can account for stratification by having a different constant term for each stratum. Let us denote  $Y_{it} \in \{0, 1\}$  the label (e.g. case status) of the  $t$ th observation of the  $i$ th stratum and  $X_{it} \in \mathbb{R}^p$  the values of the corresponding predictors. We then take the likelihood of one observation to be

$$P(Y_{it} = 1 | X_{it}) = \frac{\exp(\alpha_i + \beta^\top X_{it})}{1 + \exp(\alpha_i + \beta^\top X_{it})}$$

**Fig. 2:** Source: Wikipedia

**Stratification...**  
with respect to what?  
**By case!**

# So, which statistical model should we focus on? (Pt. 3)



## Sampled Partial likelihood:

$$\mathcal{L}^S(\beta) = \prod_{i=1}^n \frac{\exp \{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\sum_{(s,r) \in \mathcal{SR}_{t_i}} \exp \{\beta \cdot \mathbf{x}_{sr}(t_i)\}}$$

### Modeling:

## Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{\beta \cdot \mathbf{x}_{sr}(t)\}$$

### Inference:

## Conditional Logistic Regression

$$\Delta N_{sr}(t_i) | \sum_{s',r' \in \mathcal{SR}_{t_i}} \Delta N_{s'r'}(t_i) = 1 \sim \text{Categorical}(\pi_{sr}(t_i))$$

$$\pi_{sr}(t_i) = \frac{\exp \{\beta \cdot \mathbf{x}_{sr}(t_i)\}}{\sum_{(s',r') \in \mathcal{SR}_{t_i}} \exp \{\beta \cdot \mathbf{x}_{s'r'}(t_i)\}}$$



# Sampled Partial Likelihood in Practice I

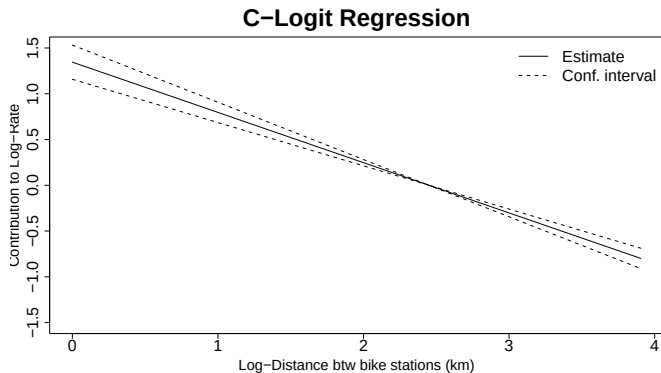
## Sampled Partial Log-Likelihood:

$$\ell^S(\beta) = \log \mathcal{L}^S(\beta) = \sum_{i=1}^n \left[ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left( \sum_{(s,r) \in \mathcal{SR}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \} \right) \right],$$

| stratum | event | sender | receiver | time      | dist     |
|---------|-------|--------|----------|-----------|----------|
| 1       | 1     | 31023  | 31261    | 0.0102023 | 2.042388 |
| 1       | 0     | 31023  | 31419    | 0.0102023 | 2.405412 |
| 1       | 0     | 32050  | 31966    | 0.0102023 | 3.271848 |
| 1       | 0     | 31246  | 31324    | 0.0102023 | 1.066778 |
| 1       | 0     | 31023  | 33200    | 0.0102023 | 2.078691 |
| 1       | 0     | 31419  | 32217    | 0.0102023 | 3.833369 |
| 2       | 1     | 31419  | 31255    | 0.0140662 | 1.683246 |
| 2       | 0     | 31071  | 31324    | 0.0140662 | 2.088897 |
| 2       | 0     | 32050  | 33200    | 0.0140662 | 2.527966 |
| 2       | 0     | 31313  | 32217    | 0.0140662 | 3.753660 |



```
rem(event ~ dist, data = sampled_dat, method = "clogit", stratum = "stratum")
```





# Who first used this approach in REMs?



## Social Networks

Volume 43, October 2015, Pages 121-135



## Relational event models for social learning in MOOCs

Duy Vu <sup>a</sup>  , Philippa Pattison <sup>b</sup>  , Garry Robins <sup>a</sup>  



Network Science

## Reliability of relational event model estimates under sampling: How to fit a relational event model to 360 million dyadic events

Published online by Cambridge University Press: 22 November 2019

Jürgen Lerner  and Alessandro Lomi

Show author details 

Article Metrics



# Towards Case-control sampling via GLM

---

Full Likelihood via Poisson Regression... At what price?

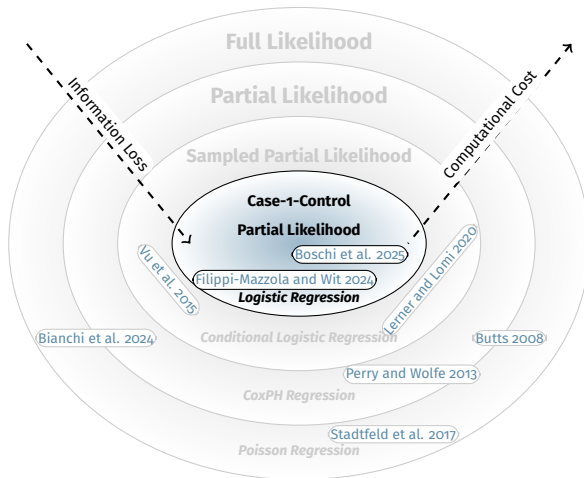
1. Strong Assumption;
2. High Computational Cost;

Well, we can do we even better!



# Case-1-control Sampling Inference

Can we do more?





# How to get a Logistic Regression... I

## Sampled Partial likelihood with $m = 1$ :

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \} + \exp \{ \beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i) \}}$$



## How to get a Logistic Regression... II

### Sampled Partial likelihood with $m = 1$ :

$$\begin{aligned}
 \mathcal{L}^S(\beta) &\stackrel{m=1}{=} \prod_{i=1}^n \frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\} + \exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}} \\
 &= \prod_{i=1}^n \frac{\frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}}{\frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}} + \frac{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}} = \prod_{i=1}^n \frac{\exp\{\beta \cdot (\mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i))\}}{1 + \exp\{\beta \cdot (\mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i))\}} \\
 &= \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta \mathbf{x}_i) \quad \Delta \mathbf{x}_i = \mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i)
 \end{aligned}$$

## So, which statistical model should we focus on? (Pt. 4)

Sampled Partial likelihood with  $m = 1$ :

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta x_i) \quad \Delta x_i = x_{s_i r_i}(t_i) - x_{s_i^* r_i^*}(t_i)$$

Modeling:

## Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{ \beta \cdot x_{sr}(t) \}$$

Inference:

## Logistic Regression

$$Y_i | \Delta x_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\pi_i)$$

$$\text{logit}(\pi_i) = \beta \cdot \Delta x_i$$

$$y_i = 1 \quad i = 1, \dots, n; \quad \text{no intercept}$$

Wondering what advantage is? **Linearity** no more!

## Case-1-control Sampling Inference via GLM in Practice I



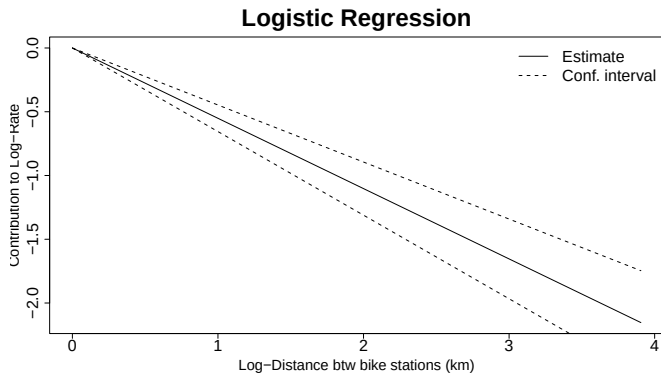
Sampled Partial likelihood with  $m = 1$ :

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta \mathbf{x}_i) \quad \Delta \mathbf{x}_i = \mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i)$$

| stratum | time      | sender_ev | receiver_ev | sender_nv | receiver_nv | dist_ev   | dist_nv   |
|---------|-----------|-----------|-------------|-----------|-------------|-----------|-----------|
| 1       | 0.0015464 | 31218     | 31255       | 31419     | 31419       | 1.6821306 | 0.0000000 |
| 2       | 0.0029034 | 31272     | 31324       | 33202     | 31966       | 1.7032925 | 2.873395  |
| 3       | 0.0096929 | 31937     | 32050       | 32050     | 31262       | 2.7475273 | 2.547647  |
| 4       | 0.0452324 | 31261     | 31262       | 31313     | 31218       | 1.1820339 | 2.201659  |
| 5       | 0.0517514 | 31272     | 31272       | 31419     | 31218       | 0.0000000 | 1.875181  |
| 6       | 0.0524939 | 33200     | 31261       | 31255     | 33202       | 0.5538851 | 1.243866  |
| 7       | 0.0531366 | 31285     | 31419       | 31419     | 33200       | 1.6052291 | 1.784903  |
| 8       | 0.0572758 | 32604     | 32217       | 31966     | 31262       | 3.0713035 | 2.821915  |
| 9       | 0.0628786 | 31285     | 33202       | 31071     | 31023       | 1.0922588 | 1.974220  |
| 10      | 0.0676353 | 31255     | 32050       | 31297     | 31272       | 2.4686079 | 1.655558  |



```
rem(~ dist, data = dat_gam, method = "gam")
```







## Hands-on $\approx$ 20 minutes

- ▶ Simulate the data in R Studio.
- ▶ Construct the appropriate datasets for different inference approaches.
- ▶ Compare the obtained results.

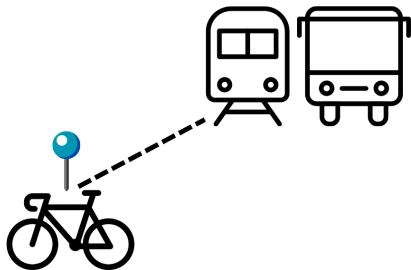
[https://martinaboschi.github.io/rem-tutorial/01-Introduction-to-REM/  
01-Simulating-REM-Data/](https://martinaboschi.github.io/rem-tutorial/01-Introduction-to-REM/01-Simulating-REM-Data/)

# All you need is linear, right?



## LET'S TALK BIKE SHARING DYNAMICS

How does distance between a bike station and a public transportation hub affect the number of bike rides? Is this effect going to be the same at 8AM or 6PM compared to 2 PM?



We need **time-varying effects**: the risk factor  $x_{sr}$  has **different effect across time**

$$\beta(t) \cdot x_{sr}(t)$$

# All you need is linear, right?



## LET'S TALK BIKE SHARING DYNAMICS

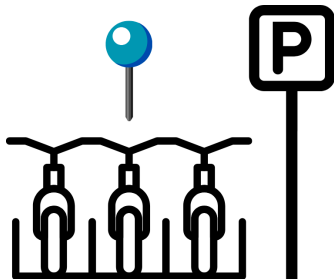
Do people bike far? What is the effect of biking 1km more if I have to bike 2kms compared to 20kms?

We need **non-linear effects**: **different effect for different values** assumed by the risk factor  $x_{sr}$

$$f(x_{sr}(t))$$



# All you need is linear, right?



## LET'S TALK BIKE SHARING DYNAMICS

Not all bike stations are the same! Some have higher/lower usage due to aspects that are unrelated to the predictors we account for (e.g. bike availability, proximity to tourist attractions...).

We need **random effects**: accounts for the **heterogeneity** arising from the system features

$$\gamma \cdot v_{sr}(t)$$

## Back to the hazard



$$\lambda_{sr}(t) = \underbrace{W_{sr}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp \left\{ \underbrace{f(x_{sr}(t), v_{sr}(t), t)}_{\text{contribution function}} \right\}$$

$$f(x_{sr}(t), v_{sr}(t), t) = \underbrace{\beta \cdot x_{sr}^{(1)}(t)}_{\text{fixed linear}} + \underbrace{\beta(t) \cdot x_{sr}^{(2)}(t)}_{\text{time-varying}} + \underbrace{f(x_{sr}^{(3)}(t))}_{\text{non-linear}} + \underbrace{\gamma \cdot v_{sr}(t)}_{\text{random}}$$

1. **fixed linear**:  $\beta$  is time-homogenous and is not a function of  $x_{sr}^{(1)}(t)$ ;
2. **time-varying**:  $\beta(t)$  is a non-linear function of time;
3. **non-linear**:  $f(x_{sr}^{(3)}(t))$  is a non-linear function of  $x_{sr}^{(3)}(t)$ ;
4. **random**:  $\gamma$  from a Gaussian random variable;



## Modeling Time-Varying Effects

- ▶ Effect of  $x_{sr}^{(2)}(t)$  as  $\rightarrow$  **Spline function of Time**

$$\beta(t) \approx \sum_{j=1}^q \theta_j b_j(t)$$

Choices to be made:

- ▶ **basis dimension**  $q$
- ▶ a **smooth basis functions**  $\{b_j\}_j$  e.g. *B-splines, thin plate regression splines,...*
- ▶ given range of variable, **basis evaluation points** (e.g. *uniform, quantile of a distribution,...*) to evaluate  $q$  **smooth non-linear functions** of time  $b_j(t)$ .

Estimating  $\beta(t) \iff$  Estimating  $\theta_j$ -s

## TVE: contribution of distance from public transport



## ► ... to log-hazard of an interaction

$$f(x_{sr}(t), v_{sr}(t), t) = \dots + \underbrace{\sum_{j=1}^q \theta_j b_j(t)}_{\text{time-varying effect}} \times \underbrace{x_{sr}^{(2)}(t)}_{\text{covariate}} + \dots$$

## ► ... to sampled partial likelihood (m=1)

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \beta(t_i)[x_{s_i r_i}^{(2)}(t_i) - x_{s_i^* r_i^*}^{(2)}(t_i)] + \dots}}{1 + e^{\dots + \beta(t_i)[x_{s_i r_i}^{(2)}(t_i) - x_{s_i^* r_i^*}^{(2)}(t_i)] + \dots}}$$

$$\downarrow$$

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}}{1 + e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}} \quad (+ \text{ smoothing penalty})$$

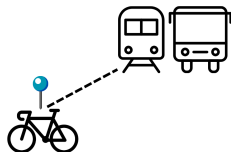
## TVE: contribution of distance from public transport



## ► ... to sampled partial likelihood (m=1)

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}}{1 + e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}} \quad (+ \text{ smoothing penalty})$$

| stratum | time      | sender_ev | receiver_ev | sender_nv | receiver_nv | distPT_ev | distPT_nv |
|---------|-----------|-----------|-------------|-----------|-------------|-----------|-----------|
| 1       | 0.0014511 | 32217     | 32604       | 32604     | 31937       | 3.8203727 | 6.2539595 |
| 2       | 0.0015844 | 31937     | 31285       | 32217     | 31246       | 1.7116404 | 3.8203727 |
| 3       | 0.0081908 | 31324     | 31218       | 31272     | 31261       | 1.3001624 | 2.2255955 |
| 4       | 0.0117318 | 31313     | 31272       | 31297     | 32604       | 2.3462775 | 0.7931991 |
| 5       | 0.0145487 | 31285     | 32217       | 31261     | 31285       | 2.3303198 | 1.4511050 |
| 6       | 0.0183787 | 33200     | 31313       | 31262     | 31419       | 4.1927036 | 1.1147747 |
| 7       | 0.0199062 | 31218     | 31285       | 31966     | 31255       | 1.9189675 | 1.8293264 |
| 8       | 0.0203390 | 31023     | 31272       | 31023     | 31324       | 0.8268842 | 0.8268842 |
| 9       | 0.0211421 | 31966     | 31246       | 33202     | 31966       | 1.8293264 | 3.2714157 |
| 10      | 0.0224279 | 33200     | 32050       | 31297     | 31071       | 4.1927036 | 0.7931991 |



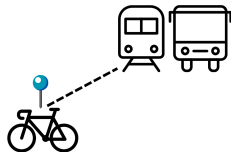
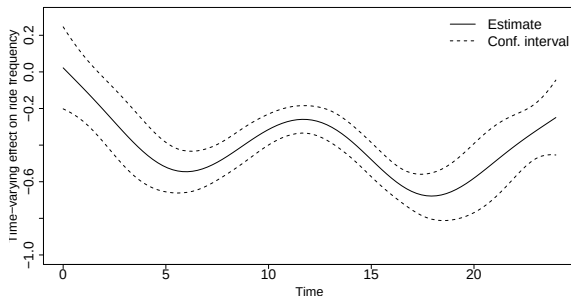
```
rem(~ tv(distPT), time = "time", data = dat_gam_tv, method = "gam")
```



## TVE: contribution of distance from public transport

► ... to **sampled partial likelihood (m=1)**

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}}{1 + e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}} \quad (+ \text{smoothing penalty})$$



# Modeling Non-Linear Effects



- ▶ Effect of  $x_{sr}^{(3)}(t)$  as  $\rightarrow$  **Spline function of Covariate**

$$f(x_{sr}^{(3)}(t)) \approx \sum_{j=1}^q \theta_j b_j(x_{sr}^{(3)}(t))$$

Choices to be made:

- ▶ **basis dimension**  $q$
- ▶ a **set of smooth basis functions**  $\{b_j\}_j$  e.g. *B-splines, thin plate regression splines,...*
- ▶ given range of variable, **basis evaluation points** (e.g. *uniform, quantile of a distribution,...*) to evaluate  $q$  **smooth non-linear functions** of covariate  $b_j(x_{sr}^{(3)}(t))$ .

Estimating  $f \iff$  Estimating  $\theta_j$ -s



## NLE: contribution of distance btw two bike stations

- ... to **log-hazard of an interaction**

$$f(\mathbf{x}_{sr}(t), t) = \dots + \underbrace{\sum_{j=1}^q \theta_j b_j(x_{sr}^{(3)}(t))}_{\text{non-linear effect}} + \dots$$

- ... to **sampled likelihood function (m=1)**

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[ \sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[ \sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$



# NLE: contribution of distance btw two bike stations

## ► ... to sampled likelihood function (m=1)

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[ \sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[ \sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$

| stratum | time      | sender_ev | receiver_ev | sender_nv | receiver_nv | dist_ev  | dist_nv  |
|---------|-----------|-----------|-------------|-----------|-------------|----------|----------|
| 1       | 0.0013772 | 33202     | 32604       | 32604     | 31937       | 2.712905 | 2.328253 |
| 2       | 0.0014538 | 31937     | 31285       | 32217     | 31246       | 1.687139 | 3.801985 |
| 3       | 0.0019790 | 33200     | 31937       | 31255     | 31261       | 1.833861 | 1.058137 |
| 4       | 0.0044957 | 31246     | 31937       | 31218     | 31937       | 1.466260 | 1.946053 |
| 5       | 0.0074578 | 31324     | 31218       | 31272     | 31261       | 1.603420 | 1.458615 |
| 6       | 0.0105801 | 31272     | 31324       | 31272     | 31937       | 1.703293 | 2.039791 |
| 7       | 0.0112535 | 31966     | 31262       | 31285     | 32217       | 2.821915 | 3.821223 |
| 8       | 0.0113760 | 33202     | 32217       | 31272     | 31261       | 3.849168 | 1.458615 |
| 9       | 0.0117538 | 31313     | 31262       | 33202     | 32604       | 2.015569 | 2.712905 |
| 10      | 0.0142903 | 31246     | 33200       | 31966     | 31255       | 1.096610 | 2.766130 |



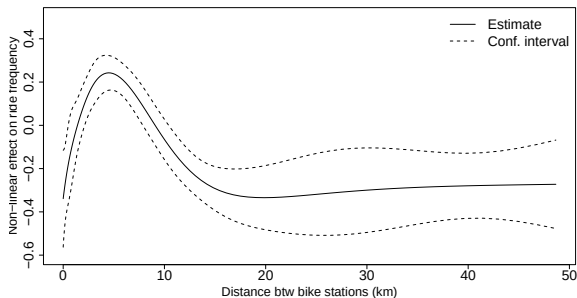
```
rem(~ nl(dist), data = dat_gam_nl, method = "gam")
```

# NLE: contribution of distance btw two bike stations



## ► ... to **sampled likelihood function (m=1)**

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[ \sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[ \sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$



# Modeling Random Effects



- ▶ Effect of  $v_{sr}(t) = \mathbb{1}\{\text{start station} = s\}$  as  $\rightarrow$  **0 - dim spline**

$$\gamma \cdot v_{sr}(t) \approx \sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s' = s\} \quad \gamma = (\gamma_1, \dots, \gamma_{|V|})$$

- ▶ Consider a model with Gaussian random effects,  $\gamma \sim \mathcal{N}(0, \sigma^2 I_{|V|})$ ;
- ▶ Same as fitting a ridge regression model (quadratic penalty term on  $\gamma$ )

Estimating  $\gamma \iff$  Estimating  $\gamma_{s'}$ -s (and  $\sigma^2$ )

## RE: contribution of bike station's popularity as start

► ... to **log-hazard of an interaction**

$$f_{sr} = \dots + \underbrace{\sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s = s'\}}_{\text{random effect}} + \dots$$

► ... to **sampld log-likelihood function (m=1)**

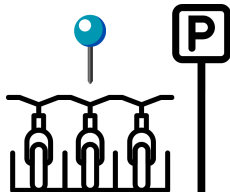
$$\ell^S(\gamma) = \log \mathcal{L}^S(\gamma) \stackrel{m=1}{=} \sum_{i=1}^n \log \left( \frac{e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}}{1 + e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}} \right) - \frac{1}{2\sigma_{\text{station}}^2} \gamma \cdot \gamma$$

## RE: contribution of bike station's popularity as start

► ... to sampled log-likelihood function ( $m=1$ )

$$\ell^S(\gamma) = \log \mathcal{L}^S(\gamma) \stackrel{m=1}{=} \sum_{i=1}^n \log \left( \frac{e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}}{1 + e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}} \right) - \frac{1}{2\sigma_{\text{station}}^2} \gamma \cdot \gamma$$

| stratum | time      | sender_ev | receiver_ev | sender_nv | receiver_nv |
|---------|-----------|-----------|-------------|-----------|-------------|
| 1       | 0.0013772 | 33202     | 32604       | 32604     | 31937       |
| 2       | 0.0014538 | 31937     | 31285       | 32217     | 31246       |
| 3       | 0.0019790 | 33200     | 31937       | 31255     | 31261       |
| 4       | 0.0044957 | 31246     | 31937       | 31218     | 31937       |
| 5       | 0.0074578 | 31324     | 31218       | 31272     | 31261       |
| 6       | 0.0105801 | 31272     | 31324       | 31272     | 31937       |
| 7       | 0.0112535 | 31966     | 31262       | 31285     | 32217       |
| 8       | 0.0113760 | 33202     | 32217       | 31272     | 31261       |
| 9       | 0.0117538 | 31313     | 31262       | 33202     | 32604       |
| 10      | 0.0142903 | 31246     | 33200       | 31966     | 31255       |



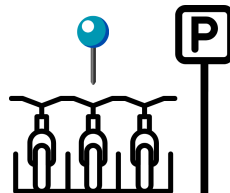
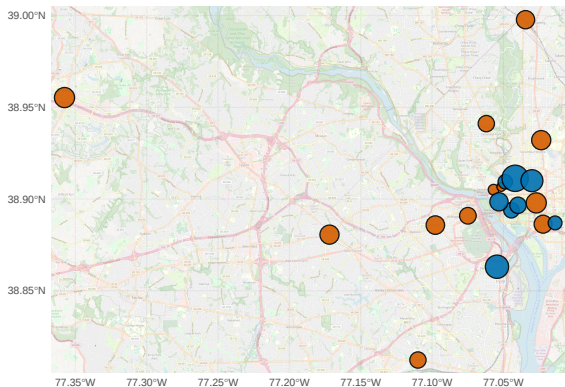
```
rem(~ re(sender), data = dat_gam_r, method = "gam")
```



## RE: contribution of bike station's popularity as start

► ... to sampled log-likelihood function ( $m=1$ )

$$\ell^S(\gamma) = \log \mathcal{L}^S(\gamma) \stackrel{m=1}{=} \sum_{i=1}^n \log \left( \frac{e^{\dots + \gamma_{s_i} - \gamma_{s_i}^* + \dots}}{1 + e^{\dots + \gamma_{s_i} - \gamma_{s_i}^* + \dots}} \right) - \frac{1}{2\sigma_{\text{station}}^2} \gamma \cdot \gamma$$





## Hands-on $\approx$ 25 minutes

- ▶ Load the data in R Studio.
- ▶ Create the case-control dataset and compute covariates.
- ▶ Fit models with linear, time-varying, non-linear, and random effects.

[https://martinaboschi.github.io/rem-tutorial/01-Introduction-to-REM/  
02-Alien-Species-Invasions/](https://martinaboschi.github.io/rem-tutorial/01-Introduction-to-REM/02-Alien-Species-Invasions/)

## Wrapping up



- ▶ (Some) Dynamic networks can be analyzed as temporal sequence of **relational events** between senders and receivers.
- ▶ REMs can capture complex temporal dynamics of interactions through:
  - both **exogenous** and **endogenous** covariates
  - flexible specification of effects (**non-linear, time-varying** and **random**, but more available...) easily estimated using efficient **logistic likelihood inference**.
- ▶ For more information on Relational Event Models, check out this tutorial paper (currently under review):



A tutorial paper on 'amore' will also be released soon..

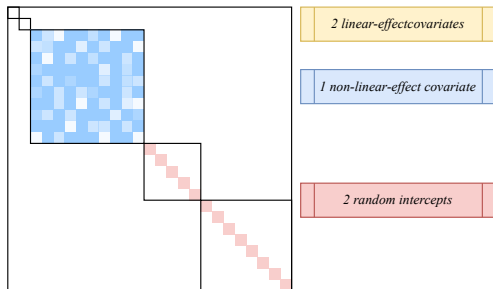


# Smooth Relational Event Model

To avoid over-fitting, we consider a *penalized partial log-likelihood*;

$$\ell^\nu(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) - P_\nu(\boldsymbol{\theta}) = \log(\mathcal{L}(\boldsymbol{\theta})) - \frac{1}{2} \sum_{l=1}^L \nu_l P_l, \quad P_l = \boldsymbol{\theta}_{j_l}^\top S_l \boldsymbol{\theta}_{j_l}, \quad j_l \subseteq \{1, \dots, |\boldsymbol{\theta}|\}$$

- ▶ **Linear:**  $S_l = 0$ ;
  - ▶ **Time-Varying / Non-Linear:**  
non-sparse  $S_l$ ;
  - ▶ **Random:**
    - **Penalized quasi-likelihood** for Gaussian random effects:
- $$\ell^\nu(\boldsymbol{\beta}, \boldsymbol{\theta}) \approx \ell(\boldsymbol{\beta}, \boldsymbol{\theta}) - \frac{1}{2} \boldsymbol{\theta}^\top \Sigma(\boldsymbol{\psi})^{-1} \boldsymbol{\theta}$$
- (Breslow and Clayton, 1993)
- $S_l$  is an identity matrix;





## References I

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